

Online Sales Dataset – Business Intelligence Analytical Report

1. Executive Summary

This report presents a business intelligence analysis of the Online Sales Dataset, focusing on sales performance, customer behavior, pricing strategy, and logistics efficiency. The objective is to demonstrate analytical reasoning, KPI definition, and insight generation aligned with Data Analyst and BI Analyst roles.

2. Dataset & Methodology

The dataset consists of multi-year transactional e-commerce records covering products, prices, discounts, shipping costs, order priorities, and geographic dimensions. A structured BI workflow was applied to ensure data reliability and analytical consistency.

Methodological steps:

- Data validation and cleaning to remove invalid or inconsistent records
- Feature engineering to derive financial and time-based metrics
- Exploratory data analysis to identify trends, patterns, and anomalies
- KPI definition aligned with business objectives

3. KPI Framework

Key performance indicators were designed to measure revenue efficiency, customer behavior, and operational performance in a business context.

- Net Revenue: Revenue after applied discounts
- Total Order Value: Net Revenue plus shipping costs
- Quantity Sold: Total number of units sold
- Return Rate: Percentage of returned orders
- Average Discount: Mean discount applied per transaction
- Shipping Cost Ratio: Shipping cost as a percentage of revenue

4. Sales & Revenue Analysis

Net revenue exceeds forty-four million dollars, supported by consistent growth in transaction volume. Time-series analysis from 2020 to 2025 reveals a stable upward trend, indicating scalable demand and low revenue volatility.

5. Product & Category Analysis

Electronics and Apparel are the primary revenue drivers, combining higher unit prices with strong demand. Secondary categories contribute smaller but stable revenue streams, suggesting opportunities for assortment and inventory optimization.

6. Geographic Performance

Revenue distribution is concentrated in a limited number of countries. This concentration enables targeted regional strategies and highlights markets with the highest return on marketing and logistics investments.

7. Customer & Channel Insights

Online transactions account for most sales volume, while in-store transactions show higher average order values. Payment behavior analysis indicates credit cards as the dominant method across channels, followed by PayPal and bank transfers.

8. Discount Impact Analysis

The average discount rate is approximately twenty-five percent. While discounts increase transaction volume, marginal revenue gains diminish beyond certain thresholds, indicating potential for more targeted discount strategies.

9. Logistics & Operations

Shipping costs remain operationally efficient, representing less than two percent of total revenue. Warehouse distribution and shipment provider performance show balanced utilization without significant bottlenecks.

10. Business Intelligence Value

This project demonstrates core BI competencies including data preparation, KPI modeling, exploratory analysis, and dashboard-driven storytelling. The integration of Python and Tableau reflects industry-standard BI workflows.

11. Conclusion

The analysis confirms strong overall business performance and identifies opportunities to improve profitability through pricing optimization, return reduction, and regional focus. The project serves as a practical example of applied data analysis and business intelligence practices.